



E80 Team 42 Project: J.A.C.K.

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E80: Experimental Engineering, SP26



Problem Statement

Climate change, pollutants, and other anthropogenic stressors have impacted global ocean surface temperatures and ultraviolet (UV) exposure levels. Understanding how these two qualities vary with ocean depth can help illustrate the changing habitat that marine life face, which is especially significant as both temperature and UV radiation influence the health of marine ecosystems. This project investigates how temperature and UV intensity changes with depth in the ocean water at Dana Point Pier, California, using an autonomous diving robot to collect data.

Impact

Many vital organisms in the marine ecosystem live at the ocean surface. Some organisms such as some fish rely on UV light for vision. Many organisms such as plankton can experience lethal harm from UV light exposure. Changes in global ocean temperature and UV intensity can be linked to anthropic causes, so illustrating the changing marine habitat can help show the severity of the impact of climate change and human activity.

Future Work

In the future, collecting data at different times of the day could be helpful in illustrating the UV penetration and warmth at higher or lower amounts of direct sunlight. Collecting data throughout the seasons in a year could help illustrate how UV and temperature from the Sun penetrates through the ocean surface changes through seasons.

Autonomous Underwater Vehicle (AUV) Features

Electrical Schematic & Sensors

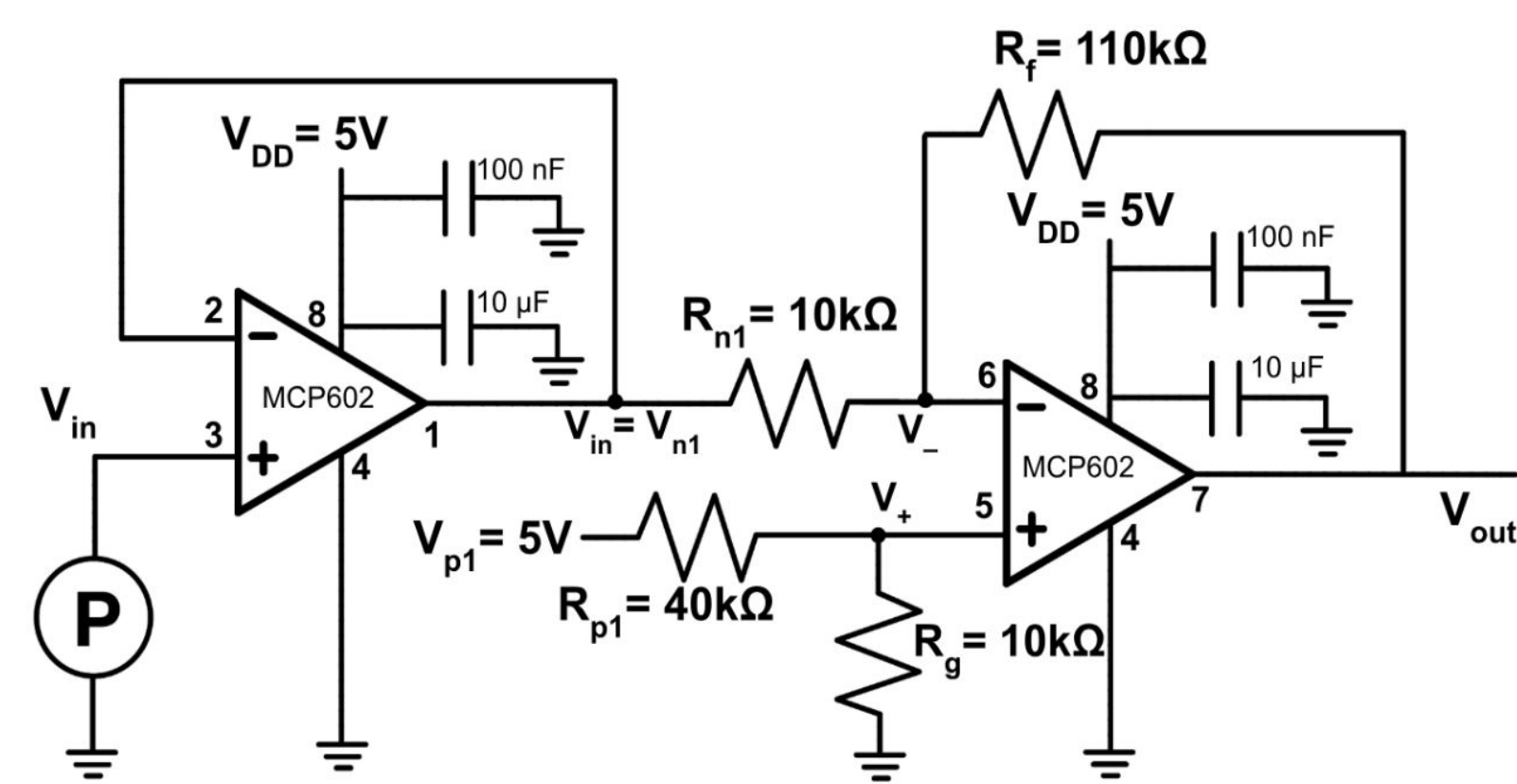


Figure 1: Pressure Sensor Interface

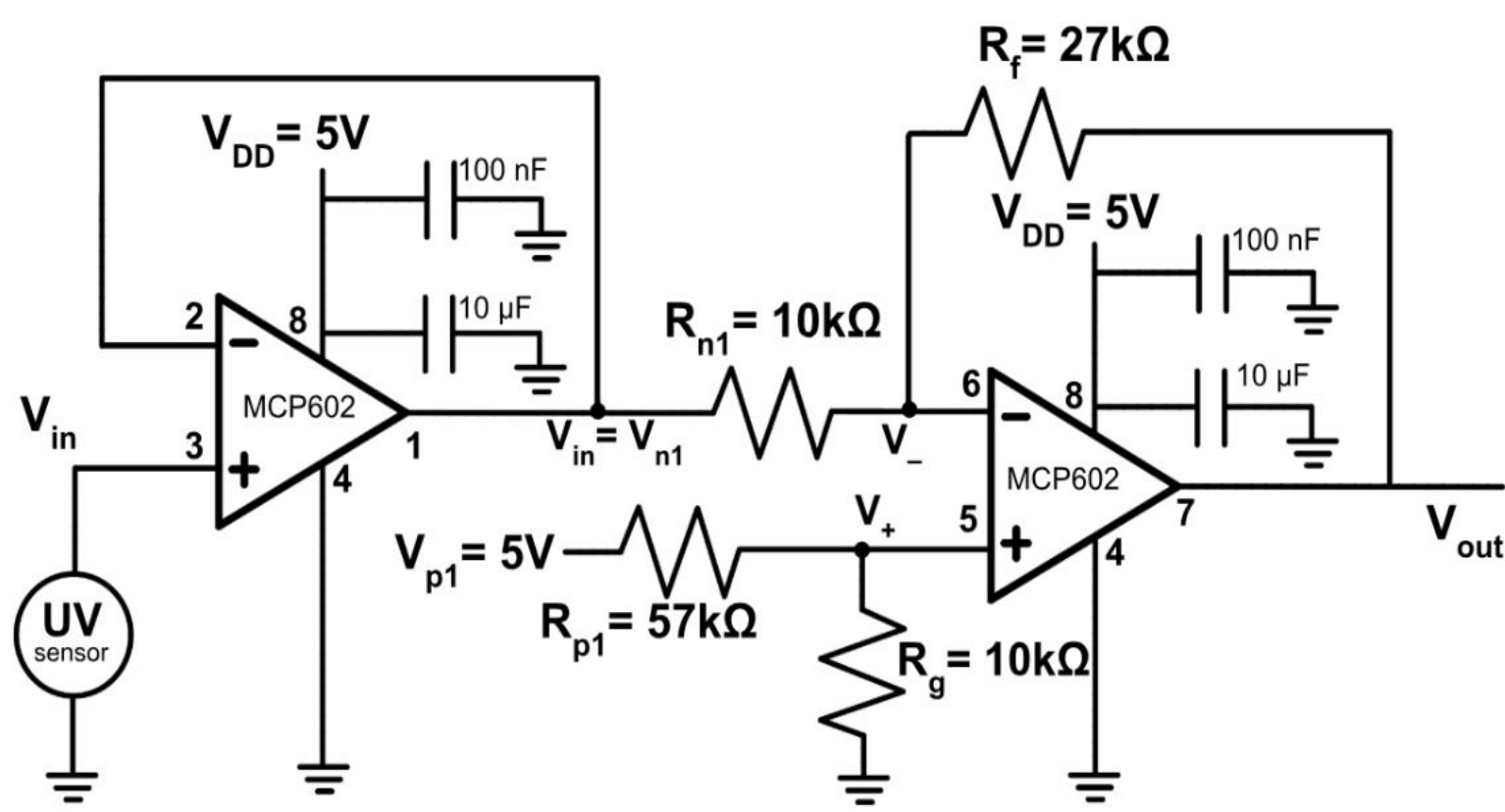


Figure 2: UV Sensor Interface

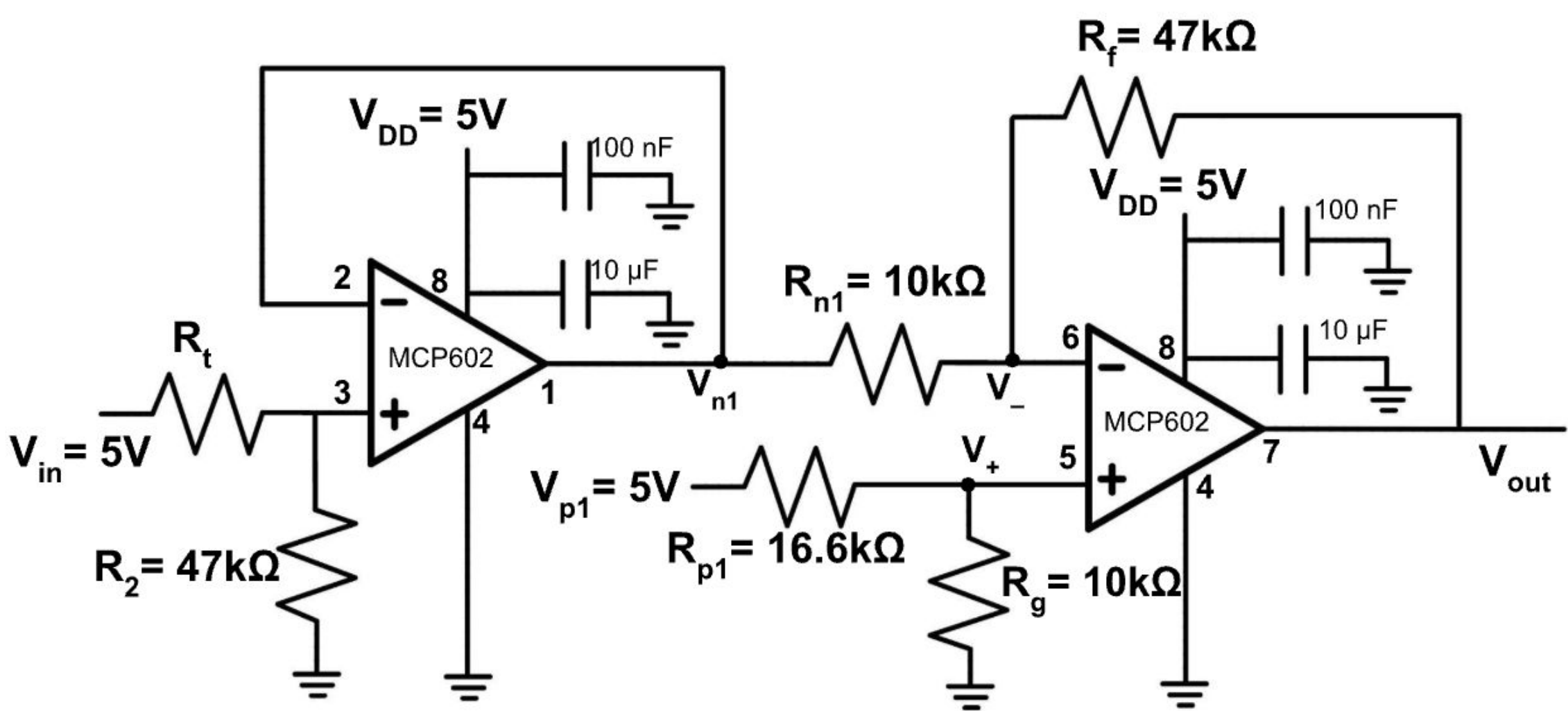
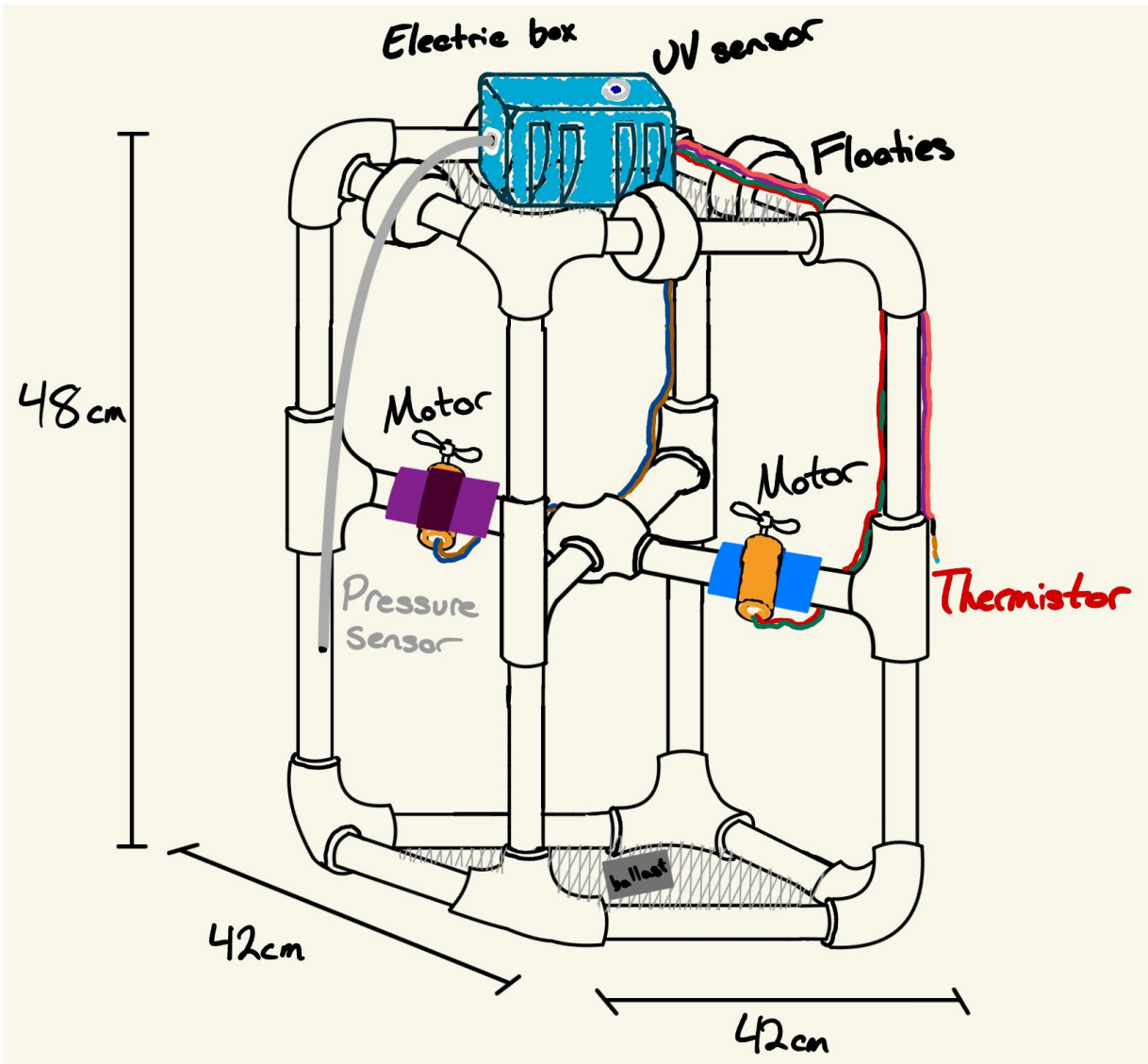
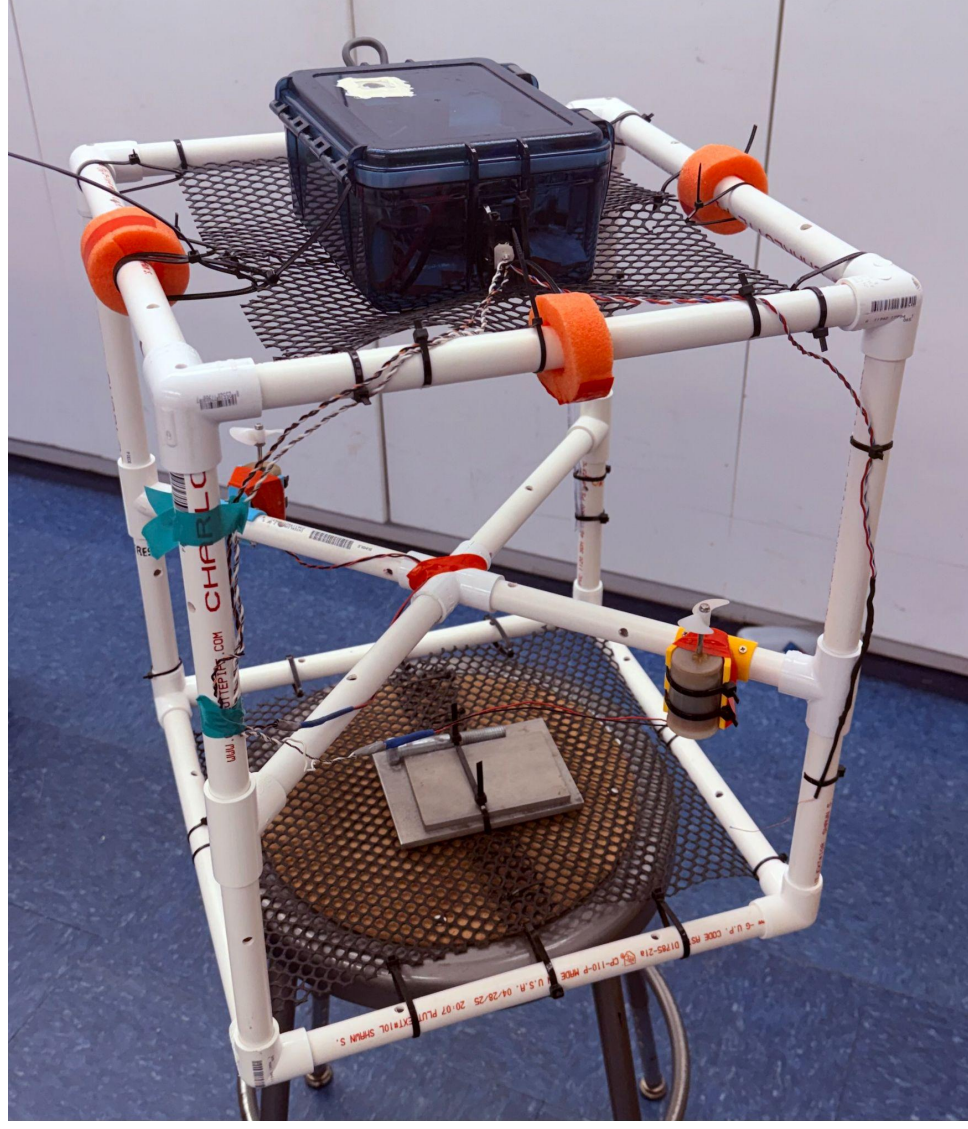


Figure 3: Thermistor Interface

Mechanical Design & Integration



AUV mechanical schematic

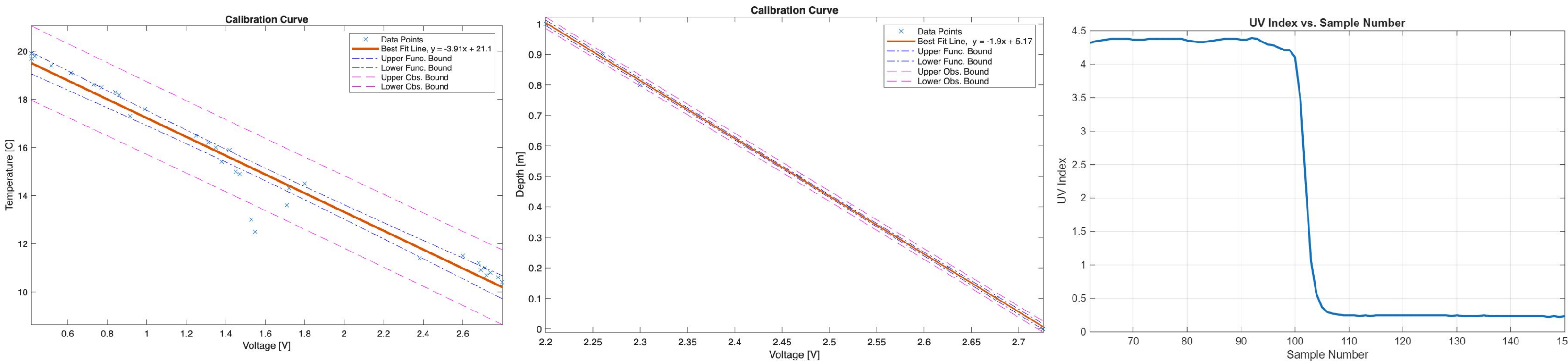


AUV deployed at Dana Point

1. Rectangular shape → Maximize stability & minimize axial tilt
2. 3 layers: Box (top), motors (middle), ballasts (bottom)
 - a. Ensure COB was positioned above COM
3. Quartz window
 - a. Minimize false UV reading
4. Motors placed outside frame
 - a. Avoid thrust force pushing upwards on electrical box

Experimental Data and Results

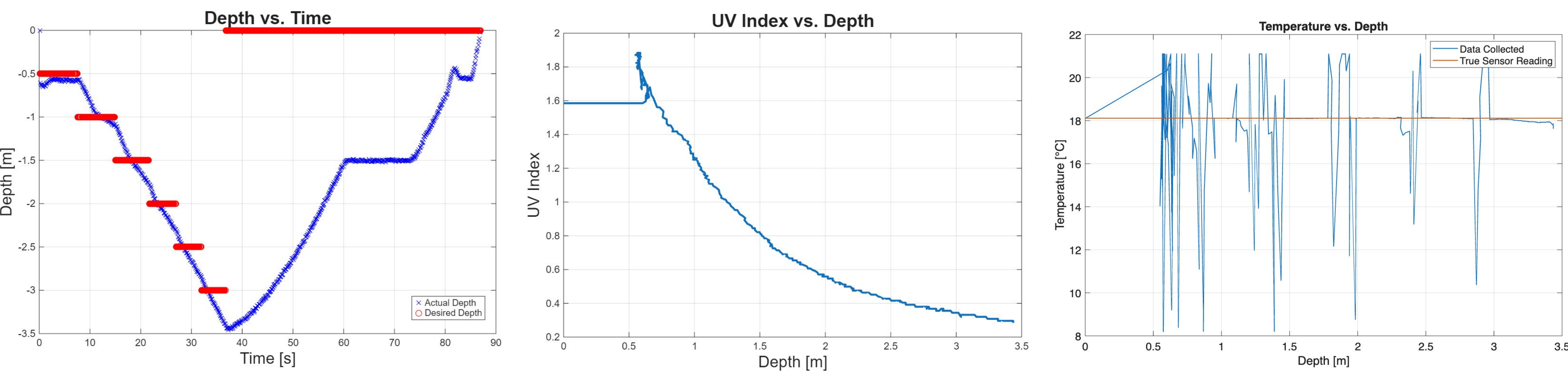
Sensor Calibration



Ground Truth Measurements

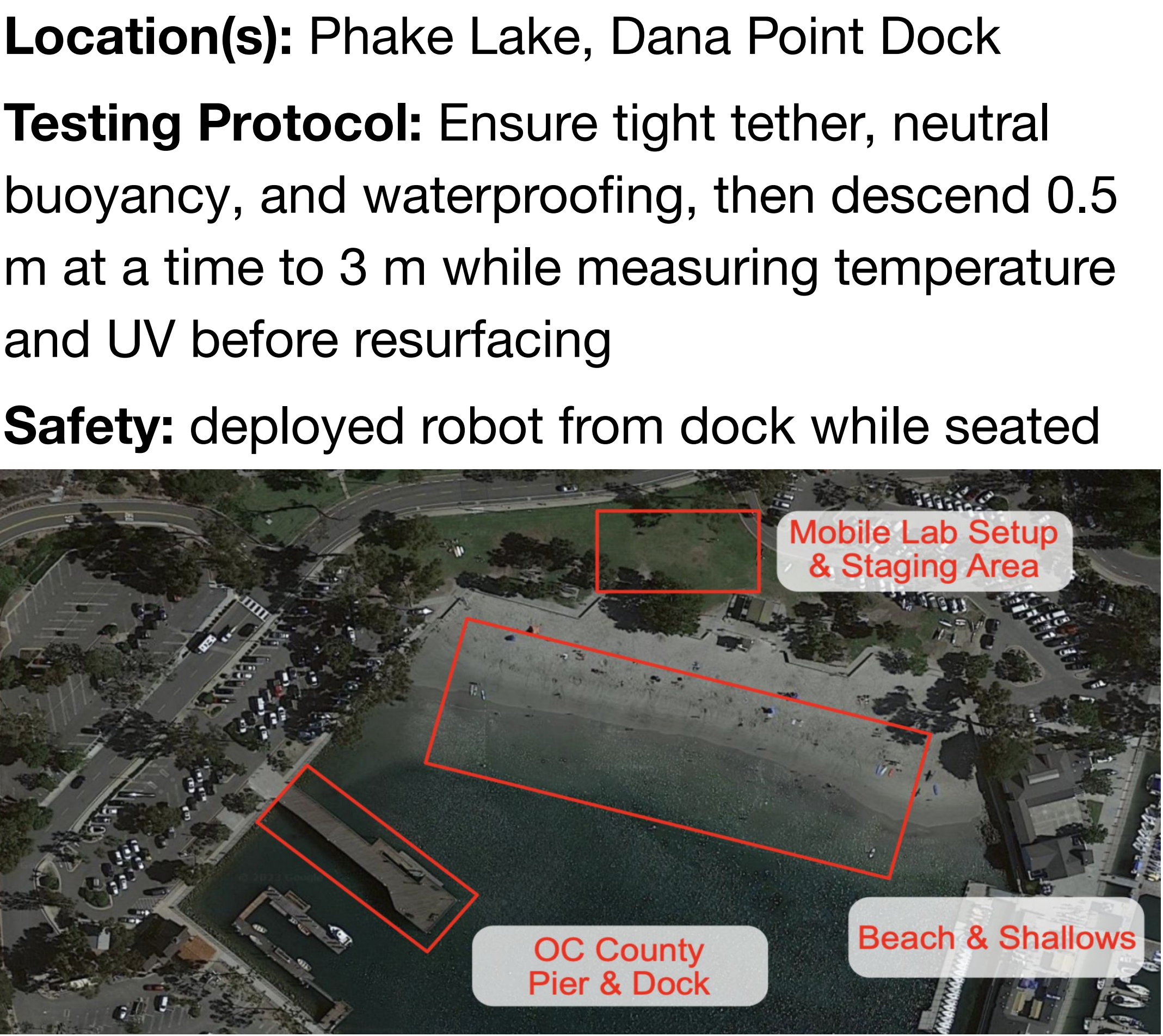
Dana Point's ocean surface temperature was 18.7 C at 8:35 AM (launch time). The pressure was 101.3 kPa. The UV intensity was around 2 above the water's surface according to the Weather Channel. The pressure calibration equation for saltwater collected at Dana Point was: $\text{Depth [m]} = -1.9 * (\text{Output_Voltage}) - 5.17$

Results & Conclusions



Above are the results of the Dana Point deployment. While temperature stayed relatively constant at 18.1 C, UV intensity decreased significantly as the robot descended to 3.5 meters. This observation shows that Dana Point's ocean water significantly attenuates UV waves. UV light can be extremely harmful to marine life, so the reach of UV waves can help inform where safe living environments are for specific UV-sensitive organisms.

Deployment Details



Location(s): Phake Lake, Dana Point Dock
Testing Protocol: Ensure tight tether, neutral buoyancy, and waterproofing, then descend 0.5 m at a time to 3 m while measuring temperature and UV before resurfacing
Safety: deployed robot from dock while seated

Acknowledgements

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